Cohen Structure Theorems

Tomàs Planelles Alonso

March 30, 2026

Abstract

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1 Topological Groups

Definition 1.1. *Given an abelian group $(G,+)$ and a topology on G we say that G is a* topological abelian group *if $+\colon G\times G\to G$ and $-\colon G\to G$ are continuous.*

Note that if $\mathcal{B}_0$ is a local basis at 0 then $T_g[\mathcal{B}_0]$ is a local basis at $g\in G$, where $T_g\colon G\to G$ is given by $T_g(h)=g+h$ which is a homeomorphism. That is, the topology of G is determined by $\mathcal{B}_0$.

Theorem 1.2. *Let H be the intersection of all neighborhoods of 0. Then H is a subgroup of G and can be described as the closure of $\{0\}$. Furthermore, G is Hausdorff if and only if $H=0$.*

Proof. By construction $0\in H$. Consider the continuous function $f\colon (g,h)\in G\times G\mapsto g-h\in G$. Given $g,h\in H$ and U a neighborhood of 0. By continuity, $f(0,0)=0$ and the product topology, there exists V and W neighborhoods of 0 such that that $V\times W\subseteq f^{-1}[U]$. Therefore $g-h=f(g,h)\in f[V\times W]\subseteq f[f^{-1}[U]]\subseteq U$. Since U was arbitrary and by the subgroup criterion we have $H\leq G$. On the other hand,

$$\begin{aligned} g\in\overline{\{0\}} &\iff 0\in U \quad \forall U \text{ neighborhood of } g\\ &\iff 0\in g+U \quad \forall U \text{ neighborhood of } 0\\ &\iff g\in H. \end{aligned}$$

So $H=\overline{\{0\}}$. If G is Hausdorff all finite sets are closed and by definition of closure $H=0$. Recall that G is Hausdorff iff Δ_G is closed. Thus, if $H=0$, since $\Delta_G=f^{-1}[\{0\}]$, G is Hausdorff. □

1.1 Topology generated by a decreasing chain

Let $(G,+)$ denote an abelian group. Consider a decreasing chain of subgroups of G,

$$G=G_0\supseteq G_1\supseteq G_2\supseteq\cdots\supseteq G_n\supseteq\cdots$$

such chain will be called *fundamental system of neighborhoods of 0.* The reason behind this name is shown in the following

Theorem 1.3. *$\mathcal{B}:=\{T_g[G_n]:n\geq 0,\, g\in G\}\subseteq G$ is a basis of a topology on G such that G becomes a topological abelian group.*

Proof. Let us first see that $\mathcal{B}$ is a basis of a topology on G. Indeed: $G=\bigcup\mathcal{B}$ and $k+G_n\subseteq T_g[G_n]\cap T_h[G_m]$ if $m\leq n$ and $k\in T_g[G_n]\cap T_h[G_m]$. To see that it is a topological abelian group it suffices to verify that $+$ and $-$ are continuous. Since $T_h(G_n)=T_g(G_n)$ if and only if $g-h\in G_n$, $T_g[G_n]+T_h[G_n]=T_{g+h}[G_n]$ and $-T_g[G_n]=T_{-g}[G_n]$ we are done. □

Note that the topological abelian subgroups G/G_n with respect the quotient topology are discrete —in fact G/H is discrete iff H is open— and can be viewed as the topology given by the sequence $(G_m/G_n)_{m\leq n}$; i.e., the image of the fundamental system of neighborhoods of 0 of G via the canonical projection.

1.2 Cauchy completion

Recall that if (X,τ) is a topological space and $(x_n)_{n\geq 0}\in X^{\mathbb{N}}$ a sequence, we say that x_n converges to $x\in X$ —written $x_n\to x$— if for every neighborhood U of x there exists $n_U\in\mathbb{N}$ such that $x_n\in U$ for $n\geq n_U$.

Definition 1.4. *Let G be a topological abelian group. A sequence $(g_n)_{n\geq 0}\in G^{\mathbb{N}}$ is a* Cauchy sequence *if for every neighborhood U of 0 there exists $n_U\in\mathbb{N}$ such that $x_n-x_m\in U$ for every $n,m\geq n_U$. We denote by $G^{\mathbb{N}}_{\mathcal{C}}$ the set of all Cauchy sequences of G.*

Define an operation in $G_{\mathcal{C}}^{\mathbb{N}}$ component wise and we get an abelian group structure since the sum of two Cauchy sequences is a Cauchy sequence.

Definition 1.5. *Two Cauchy sequences* $(g_n)_{n\geq 0}$, $(h_n)_{n\geq 0} \in G_{\mathcal{C}}^{\mathbb{N}}$ *are* equivalent *if* $g_n - h_n \to 0$ *in* G.

It is clear that equivalence class $[(0)_{n\geq 0}]$ is a subgroup of $G_{\mathcal{C}}^{\mathbb{N}}$. We denote by $\hat{G} := G_{\mathcal{C}}^{\mathbb{N}}/[(0)_{n\geq 0}]$ the quotient abelian group. For each $g \in G$ we can define a constant sequence $(g)_{n\geq 0} \in G_{\mathcal{C}}^{\mathbb{N}}$ and take it's equivalence class given rise to a homomorphism

$$\begin{array}{rccc} \phi\colon & G & \to & \hat{G} \\ & g & \mapsto & [(g)_{n\geq 0}]. \end{array}$$

Theorem 1.6. *ϕ is one-to-one if and only if G is Hausdorff.*

Proof. Since $G \hookrightarrow G_{\mathcal{C}}^{\mathbb{N}}$ we only need to check the homomorphism $p\colon G_{\mathcal{C}}^{\mathbb{N}} \twoheadrightarrow \hat{G}$. It is clear that $\ker p = [(0)_{n\geq 0}]$. Hence, by theorem 1.2,

$$\phi \text{ is one-to-one} \iff [(0)_{n\geq 0}] = 0 \iff \overline{\{0\}} = \{0\} \iff G \text{ is Hausdorff.}$$

□

If the topology of G is given by a fundamental system of neighborhoods of 0, $(G_n)_{n\geq 0}$ then we can endow $\hat{G}$ with a topology giving by the decreasing chain

$$\hat{G} = \hat{G}_0 \supseteq \hat{G}_1 \supseteq \hat{G}_2 \supseteq \cdots \supseteq \hat{G}_n \supseteq \cdots$$

where $\hat{G}_n := \phi[G_n]$; i.e., the equivalence class of sequences that eventually fall inside G_n. It is easy to see that $\hat{G}$ is a topological abelian group and ϕ is continuous with respect this choice of topology. By theorem 1.2 and the definition of $\hat{G}_n$, since

$$\bigcap_{U \text{ neighborhood of } 0} U = \bigcap_{n\geq 0} \hat{G}_n = 0$$

we deduce that $\hat{G}$ is Hausdorff.

To finish this subsection we are going to see the functoriality properties of $\hat{}$. Let $f\colon G \to H$ be a continuous homomorphism where G and H are topological abelian groups. Given a Cauchy sequence $(g_n)_{n\geq 0}$, consider the sequence $(f(g_n))_{n\geq 0}$. Let U be any neighborhood of 0 of H. Since f is continuous and $f(0) = 0$, $f^{-1}[U]$ is a neighborhood of 0 in G. Thus there exists an integer n_U such that $g_n - g_m \in f^{-1}[U]$ for all $n, m \geq n_U$ so

$$f(g_n) - f(g_m) = f(g_n - g_m) \in f[f^{-1}[U]] \subseteq U$$

for all $n, m \geq n_U$. Hence f induces a homomorphism $\hat{f}\colon \hat{G} \to \hat{H}$. It is easy to check that $\widehat{f \circ g} = \hat{f} \circ \hat{g}$ and the following diagram commutes

$$\begin{tikzcd} G \arrow[r, "f"] \arrow[d, "\phi_G"'] & H \arrow[d, "\phi_H"] \\ \hat{G} \arrow[r, "\hat{f}"] & \hat{H}. \end{tikzcd}$$

1.3 Inverse limit as an algebraic completion

Definition 1.7. *An* inverse limit *is a limit indexed by the category ω^{op}; i.e., the terminal cone of a diagram*

$$\cdots \xrightarrow{\theta_4} G_3 \xrightarrow{\theta_3} G_2 \xrightarrow{\theta_2} G_1 \xrightarrow{\theta_1} G_0 \tag{1}$$

which is denoted as $\varprojlim G_n$:

$$\begin{tikzcd}
\varprojlim G_n \arrow[d, "\cdots"'] \arrow[dr, "\pi_3"] \arrow[drr, "\pi_2"] \arrow[drrr, "\pi_1"] \arrow[drrrr, "\pi_0"] & & & & \\
\cdots \arrow[r, "\theta_4"'] & G_3 \arrow[r, "\theta_3"'] & G_2 \arrow[r, "\theta_2"'] & G_1 \arrow[r, "\theta_1"'] & G_0
\end{tikzcd}$$

We will only consider diagrams $F\colon \omega^{\mathrm{op}} \to \mathbf{Ab}$ called *inverse systems*, which consist of a sequence $(G_n)_{n\geq 0}$ of abelian groups and homomorphisms $\theta_{n+1}\colon G_{n+1} \to G_n$ for every n. If each θ_n is surjective the inverse system is called *surjective*. Note that, in this case, π_n —defined above— are also surjective.

Theorem 1.8. *We have*

$$\varprojlim G_n = \left\{ (g_n)_{n\geq 0} \in \prod_{n\geq 0} G_n : \theta_{n+1}(g_{n+1}) = g_n \ \ \forall n \right\}.$$

Proof. Let us denote $\widetilde{G}$ the set defined on the right hand side. If A is a cone of the diagram 1; i.e.,

$$\begin{tikzcd}
A \arrow[d, "\cdots"'] \arrow[dr, "\mu_3"] \arrow[drr, "\mu_2"] \arrow[drrr, "\mu_1"] \arrow[drrrr, "\mu_0"] & & & & \\
\cdots \arrow[r, "\theta_4"'] & G_3 \arrow[r, "\theta_3"'] & G_2 \arrow[r, "\theta_2"'] & G_1 \arrow[r, "\theta_1"'] & G_0
\end{tikzcd}$$

Then the homomorphism $f\colon a \in A \mapsto (\mu_n(a))_{n\geq 0} \in \widetilde{G}$ is unique by extensionality and makes $\widetilde{G}$ a terminal cone of (1). □

Definition 1.9. *A sequence $(g_n)_{n\geq 0}$ is* coherent *if $(g_n)_{n\geq 0} \in \varprojlim G_n$.*

Let's study the case where G is a topological abelian group given by the sequence $(G_n)_{n\geq 0}$ and consider the inverse limit of the inverse system[1]

$$\cdots \xrightarrow{p_{n+2}} G/G_{n+1} \xrightarrow{p_{n+1}} G/G_n \xrightarrow{p_n} \cdots \xrightarrow{p_3} G/G_2 \xrightarrow{p_2} G/G_1 \xrightarrow{p_1} G$$

We endow $\varprojlim G/G_n$ with the initial topology generated by $\{\pi_m\colon \varprojlim G/G_n \to G/G_m\}_{m\geq 0}$ —the topology generated by the subbase $\{\pi_n^{-1}[U] : U \text{ is open in } G/G_n\}_{n\geq 0}$— so that it becomes a topological abelian group[2].

1.4 Cauchy and inverse limit comparison

We would like to compare $\varprojlim G/G_n$ with $\hat{G}$. It is clear that given $(g_n)_{n\geq 0} \in G_{\mathcal{C}}^{\mathbb{N}}$ then its image

Maybe write it with theorems instead of text

[1] Note that $G/G_n \cong (G/G_{n+1})/(G_n/G_{n+1})$.

[2] It is not a "straight-forward-proof" but it will be done using an homeomorphic isomorphism with the topological abelian group $\hat{G}$.

under $p\colon G \to G/G_m$ is ultimately constant; i.e., given the neighborhood G_m since $(g_n)_{n\geq 0}$ is Cauchy, there exists an $n_m \in \mathbb{N}$ such that $g_k - g_l \in G_m$ for $k, l \geq n_m$ that is $g_k + G_m = g_l + G_m$. Let's denote this constant by $\gamma_m^{(g_n)_{n\geq 0}} \in G/G_m$—or γ_m when $(g_n)_{n\geq 0}$ is clear from context—. Since $G_{m+1} \leq G_m$ the sequence $(\gamma_m)_{m\geq 0}$ is coherent. By the definition of equivalence of Cauchy sequences the choice of γ_m is well defined in $\hat{G}$. Hence we have a map

$$\begin{array}{rccc} \varphi\colon & \hat{G} & \hookrightarrow & \varprojlim G/G_n \\ & [(g_n)_{n\geq 0}] & \longmapsto & (\gamma_m^{(g_n)_{n\geq 0}})_{m\geq 0}. \end{array}$$

Conversely, given $(\gamma_n)_{n\geq 0} \in \varprojlim G/G_n$ we can choose a $g_n \in G$ for each $n \geq 0$ such that $\gamma_n = g_n + G_n$. By the definition of coherence and theorem 1.8 we have $g_{n+1} + G_n = g_n + G_n$ therefore, $(g_n)_{n\geq 0} \in G_{\mathcal{C}}^{\mathbb{N}}$. Thus, $\varphi([(g_n)_{n\geq 0}]) = (\gamma_n)_{n\geq 0}$, concluding that φ is an isomorphism. It is easy to see that for any map π_n the composition $\pi_n \circ \varphi$ is constant hence continuous, by the universal property of the initial topology on $\varprojlim G/G_n$, implies that φ is continuous. On the other hand, the image of $\hat{G}_m$ under φ are the sets

$$\{(\gamma_n)_{n\geq 0} \in \varprojlim G/G_n : \gamma_j = 0 \quad \forall j \leq m\} = \bigcap_{i=0}^{m} \pi_i^{-1}[\{0\}],$$

so φ is a homeomorphic isomorphism of groups, $\hat{G} \cong \varprojlim G/G_n$. From now on we will denote by $\hat{G}$ either $\hat{G}$ or $\varprojlim G/G_n$ and we will use both the analytical and algebraic description.

For example, it is easy to see that $\hat{G}_n = \ker(\hat{G} \to G/G_n)$.

1.5 Completeness

Recall that

Definition 1.10. *Given a topological abelian group G, we say that it is complete if all Cauchy sequences converge.*

Let $(g_n)_{n\geq 0}$ be a convergent sequence to $g \in G$; i.e., for all neighborhood of 0, U, there exists an $n_U \geq 0$ such that for all $n \geq n_U$ we have $g_n \in g - U$. That is, $g - g_n \in U$. Since G is a topological group we have that $(g_n)_{n\geq 0} \in G_{\mathcal{C}}^{\mathbb{N}}$. So in a complete topological abelian group all Cauchy sequences converge and all convergent sequences are Cauchy.

We end this section giving an alternative definition of completeness of Hausdorff topological abelian groups and proving that the Hausdorff abelian group $\hat{G}$ is complete.

Theorem 1.11. *Given a topological abelian group G. The homomorphism $\phi\colon G \to \hat{G}$ is an isomorphism if and only if G is Hausdorff and complete.*

Proof. We already know that ϕ is one-to-one iff G is Hausdorff. On the other hand, ϕ is onto iff every Cauchy sequence is equivalent to a constant sequence iff every sequence converges iff G is complete. □

From now on the algebraic description of $\hat{G}$ will come in handy to prove its completeness.

Let $(A_n)_{n\geq 0}$ be an inverse system and $A := \prod_{n\geq 0} A_n$. The map

$$\begin{array}{rccc} d^A\colon & A & \longrightarrow & A \\ & (a_n)_{n\geq 0} & \longmapsto & (a_n - \theta_{n+1}(a_{n+1}))_{n\geq 0} \end{array}$$

is a homomorphism of abelian groups such that $\ker d^A = \varprojlim A_n$. If each θ_n is surjective; i.e., $(A_n)_{n\geq 0}$ is surjective, then d^A is also surjective.

Definition 1.12. *Given $(A_n)_{n\geq 0}$, $(B_n)_{n\geq 0}$ and $(C_n)_{n\geq 0}$ three inverse systems. If for each $n \geq 0$ we have the following commutative diagram of exact sequences*

$$\begin{tikzcd}
0 \arrow[r] & A_{n+1} \arrow[r] \arrow[d] & B_{n+1} \arrow[r] \arrow[d] & C_{n+1} \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & A_n \arrow[r] & B_n \arrow[r] & C_n \arrow[r] & 0,
\end{tikzcd}$$

we say that we have an exact sequence of inverse systems *denoted by*

$$0 \longrightarrow (A_n)_{n\geq 0} \longrightarrow (B_n)_{n\geq 0} \longrightarrow (C_n)_{n\geq 0} \longrightarrow 0.$$

Proposition 1.13. *Let $0 \to (A_n)_{n\geq 0} \to (B_n)_{n\geq 0} \to (C_n)_{n\geq 0} \to 0$ be an exact sequence of inverse systems then*

$$0 \longrightarrow \varprojlim A_n \longrightarrow \varprojlim B_n \longrightarrow \varprojlim C_n$$

is exact. Also, if $(A_n)_{n\geq 0}$ is a surjective system,

$$0 \longrightarrow \varprojlim A_n \longrightarrow \varprojlim B_n \longrightarrow \varprojlim C_n \longrightarrow 0$$

is exact.

Proof. We have the following commutative diagram of exact sequences

$$\begin{tikzcd}
0 \arrow[r] & \prod_{n\geq 0} A_n \arrow[r] \arrow[d, "d^A"] & \prod_{n\geq 0} B_n \arrow[r] \arrow[d, "d^B"] & \prod_{n\geq 0} C_n \arrow[r] \arrow[d, "d^C"] & 0 \\
0 \arrow[r] & \prod_{n\geq 0} A_n \arrow[r] & \prod_{n\geq 0} B_n \arrow[r] & \prod_{n\geq 0} C_n \arrow[r] & 0.
\end{tikzcd}$$

Thus, by the snake lemma, we get an exact sequence

$$\begin{tikzcd}
0 \arrow[r] & \ker d^A \arrow[r] & \ker d^B \arrow[r] & \ker d^C \arrow[dll, rounded corners] & \\
& \operatorname{coker} d^A \arrow[r] & \operatorname{coker} d^B \arrow[r] & \operatorname{coker} d^C \arrow[r] & 0.
\end{tikzcd}$$

□

Corollary 1.14. *Let G be a topological abelian group with topology defined by a sequence $(G_n)_{n\geq 0}$. If $0 \to H \xrightarrow{\iota} G \xrightarrow{\pi} K \to 0$ is exact then*

$$0 \longrightarrow \hat{H} \longrightarrow \hat{G} \longrightarrow \hat{K} \longrightarrow 0$$

is exact, where the topologies of H and K are given by the sequences $(\iota^{-1}[G_n])_{n\geq 0}$ and $(\pi[G_n])_{n\geq 0}$ respectively.

Proof. It is clear that $(H/\iota^{-1}[G_n])_{n\geq 0}$ is a surjective inverse system and

$$0 \longrightarrow (H/\iota^{-1}[G_n])_{n\geq 0} \longrightarrow (G/G_n)_{n\geq 0} \longrightarrow (K/\pi[G_n])_{n\geq 0} \longrightarrow 0$$

is exact. Now apply proposition 1.13. □

Recall that G/G_n has the discrete topology as a topological abelian subgroup of G. Hence G/G_n is Hausdorff —so $\phi\colon G/G_n \hookrightarrow \widehat{G/G_n}$— and every Cauchy sequence is ultimately constant in G/G_n so that ϕ is onto. Thus $\widehat{G/G_n} \cong G/G_n$. Applying corollary 1.14 for $H = G_n$ and $K = G/G_n$ we get

$$\hat{G}/\hat{G}_n \cong \widehat{G/G_n} \cong G/G_n.$$

Taking inverse limits this becomes

Corollary 1.15. $\hat{\hat{G}} \cong \hat{G}$.

So $\hat{G}$ is Hausdorff and complete: $\phi\colon \hat{G} \to \hat{\hat{G}}$ is one-to-one and onto and apply theorem 1.11.

2 Filtrations

All rings will be commutative with an identity element.

Definition 2.1. *Given a ring $(A, +, \cdot)$ and a topology on A, we say that A is a* topological ring *if $(A, +)$ is a topological group and $\cdot\colon A \times A \to A$ is continuous.*

We will study topologies generated by an ideal of a ring A,

Definition 2.2. *Let A be a ring and $\mathfrak{a} \leq A$ and ideal. We will call* $\mathfrak{a}$-adic topology *on A the topology given by the fundamental system of neighborhoods of* 0 $(\mathfrak{a}^n)_{n\geq 0}$.

Proposition 2.3. *A with the $\mathfrak{a}$-adic topology is a topological ring.*

Proof. We only need to check that $\cdot$ is continuous. We can reduce it to open sets of the form $ab + \mathfrak{a}^n$ since $(A, +)$ is a topological abelian group. By the definition of an ideal we have

$$(a + \mathfrak{a}^n)(b + \mathfrak{a}^n) \subseteq ab + \mathfrak{a}^n$$

so $\cdot$ is continuous. □

We know that the product of two Cauchy sequences is Cauchy and the product of a Cauchy sequence by a sequence that converges to 0 converges to 0. Therefore $A^{\mathbb{N}}_{\mathcal{C}}$ is a ring and $[(0)_{n\geq 0}]$ is an ideal and so $\hat{A}$ is a ring. Since the ideals $(\widehat{\mathfrak{a}^n})_{n\geq 0}$ are a fundamental systems of neighborhoods of 0 we can view $\hat{A}$ as a topological ring. However it is not always true that $\widehat{\mathfrak{a}^n} = \hat{\mathfrak{a}}^n$ so the initial topology on $\hat{A}$ may not be its $\hat{\mathfrak{a}}$-topology. In particular, $\hat{A}$ may not be complete with respect to its $\hat{\mathfrak{a}}$-topology.

Definition 2.4. *Let M be an A-module such that M is a topological space. We say that M is a* topological A-module *if $(M, +)$ is a topological abelian group and $\cdot\colon A \times M \to M$ is continuous.*

Likewise, we can check that the topology generated by the decreasing sequence of subgroups $(\mathfrak{a}^n M)_{n\geq 0}$ gives M the structure of a topological A-module. Since $M/\mathfrak{a}^n M$ is an $A/\mathfrak{a}^n$-module and there is a canonical projection $\theta_{n+1}\colon A/\mathfrak{a}^{n+1} \twoheadrightarrow A/\mathfrak{a}^n$, we have a canonical homomorphism of modules

$$\alpha_{n+1}\colon M/\mathfrak{a}^{n+1}M \twoheadrightarrow M/\mathfrak{a}^n M$$

i.e., $\alpha_{n+1}(\lambda a) = \theta_{n+1}(\lambda)\alpha_{n+1}(a)$ where $\lambda \in A/\mathfrak{a}^{n+1}$ and $a \in M/\mathfrak{a}^{n+1}M$. With this in mind the completion $\hat{M}$ becomes an $\hat{A}$-module which is, in fact, a topological $\hat{A}$-module.

We wish to compare other topologies of M generated by a decreasing chain of submodules. We will need the following definition;

Definition 2.5. *Let M be an A-module. A decreasing sequence of submodules*

$$M = M_0 \supseteq M_1 \supseteq \cdots \supseteq M_n \supseteq \cdots$$

is called a filtration *of M.*

Let $\mathfrak{a}$ be an ideal of A. A filtration $(M_n)_{n\geq 0}$ of M is called an $\mathfrak{a}$-filtration *if $\mathfrak{a}M_n \subseteq M_{n+1}$ for all $n \geq 0$, and a* stable $\mathfrak{a}$-filtration *if there exists $n_0 \in \mathbb{N}$ such that the equality holds for all $n \geq n_0$.*

We will see that all stable $\mathfrak{a}$-filtrations define the same topology on M:

Lemma 2.6. *If $(M_n)_{n\geq 0}$ is a stable $\mathfrak{a}$-filtration of M there exists an integer $n_0 \in \mathbb{N}$ such that $M_{n+n_0} \subseteq \mathfrak{a}^n M$ and $\mathfrak{a}^{n+n_0} M \subseteq M_n$ for all $n \geq 0$. We call the topology generated by any stable $\mathfrak{a}$-filtration of M the* $\mathfrak{a}$-topology.

Proof. Since $\mathfrak{a}M_n \subseteq M_{n+1}$ we have $\mathfrak{a}^n M \subseteq M_{n+1}$. Let $n_0 \in \mathbb{N}$ be such that $\mathfrak{a}M_n = M_{n+1}$ for all $n \geq n_0$, hence $M_{n+n_0} = \mathfrak{a}^n M_{n_0} \subseteq \mathfrak{a}^n M$ for all $n \geq 0$. □

3 Graded Rings and Modules

In our particular case we would like to know which conditions must hold so that $M \hookrightarrow \hat{M}$ —equivalently, when is the $\mathfrak{a}$-topology Hausdorff— as well as prove some properties of $\hat{A}$ and $\hat{M}$ —in fact any property of $\hat{M}$ will hold as $\hat{A}$ is the completion of the A-module A—. To do so, the notion of *graded ring* and *graded A-module* comes in handy.

Definition 3.1. *A* graded ring *is a tuple $(A, (A_n)_{n\geq 0})$ where A is a ring and $(A_n)_{n\geq 0}$ a family of —additive— subgroups such that*

$$A = \bigoplus_{n\geq 0} A_n$$

and $A_m A_n \subseteq A_{n+m}$ for all $n, m \geq 0$. We will denote

$$A_+ := \bigoplus_{n>0} A_n.$$

It is clear that A_0 is a ring, A_n are A_0-modules and A_+ is an ideal of A.

Similarly,

Definition 3.2. *Let $(A, (A_n)_{n\geq 0})$ be a graded ring. A* graded A-module *is an A-module M together with a family of —additive– subgroups $(M_n)_{n\geq 0}$ such that $A_n M_m \subseteq M_{m+n}$ for all $m, n \geq 0$ and*

$$M = \bigoplus_{n\geq 0} M_n. \tag{2}$$

If M is a graded A-module, we say that $m \in M$ is homogeneous of degree n *if $m \in M_n$. By (2) any $m \in M$ can be written uniquely as a finite sum $\sum_{n\geq 0} m_n$ where $m_n \in M_n$. The non-zero components m_n are called* homogeneous components *of m.*

Proposition 3.3. *Let A be a graded ring, then A is a Noetherian ring if and only if A_0 is a Noetherian ring and A is finitely generated as an A_0-algebra.*

Proof. The map $A \twoheadrightarrow A_0$ has kernel A_+ thus $A_0 \cong A/A_+$ so A_0 is Noetherian. Say $A_+ = (a_1, \ldots, a_r)$ we can also take each a_i as a homogeneous element of degree k_i. It is clear that $A_0 \subseteq A_0[a_1, \ldots, a_r]$. Given n fixed and supposing that $A_k \subseteq A_0[a_1, \ldots, a_r]$ for all $k < n$ we see that each $y \in A_n$ can be written as $y = \sum_{i=1}^r y_i a_i$ where each $y_i \in A_{n-k_i}$. By supposition we get $y \in A_0[a_1, \ldots, a_r]$. Therefore, by induction, $A = A_0[a_1, \ldots, a_r]$ is a finitely generated A_0-algebra. The converse holds by Hilbert's basis theorem. □ Appendix

3.1 Equivalence of topologies

It is clear that given an A-module M with the $\mathfrak{a}$-topology and an A-submodule N then the $\mathfrak{a}$-topology on M/N is equivalent to the topology induced on M/N, since

$$\frac{\mathfrak{a}^n M + N}{N} = \mathfrak{a}^n \frac{M+N}{M}.$$

Similarly given to A-modules M and N with the $\mathfrak{a}$-topology we have

$$\mathfrak{a}^n(M \oplus N) = \mathfrak{a}^n M \oplus \mathfrak{a}^n N$$

so the $\mathfrak{a}$-topology on $M \oplus N$ is equal to the product of both topologies. In fact, recall that given an exact sequence of A-modules $0 \to N \to M \xrightarrow{\varphi} L \to 0$ we have $M \cong N \oplus L$ —i.e. the sequence is *split*— if and only if there exists an A-module homomorphism $\psi\colon L \to M$ such that $\varphi \circ \psi = \mathrm{id}_L$. From this follows that

$$\widehat{M \oplus N} \cong \hat{M} \oplus \hat{N} \tag{3}$$

by functoriality of $\hat{}$ and corollary 1.14 applied to the trivial extension of M by N.

What's not trivial is that the $\mathfrak{a}$-topology of an A-submodule N equals the $\mathfrak{a}$-topology on M restricted to N.

Given an ideal $\mathfrak{a}$ of a ring A and an $\mathfrak{a}$-filtration $(M_n)_{n\geq 0}$ of an A-module M we define the graded ring

$$A^* := \bigoplus_{n\geq 0} \mathfrak{a}^n$$

and the graded A^*-module

$$M^* := \bigoplus_{n\geq 0} M_n.$$

If $\mathfrak{a}$ is finitely generated, say by $a_1, \dots, a_r$, then $A^* = A[a_1, \dots, a_r]$. Thus if A is Noetherian, A^* is also Noetherian by Hilbert's basis theorem.

Lemma 3.4. *Let A be a Noetherian ring, M a finitely generated A-module, $(M_n)_{n\geq 0}$ an $\mathfrak{a}$-filtration. Then, M^* is a finitely generated A^*-module if and only if the filtration $(M_n)_{n\geq 0}$ is stable.*

Proof. Consider de A^*-submodules

$$M_n^* := M_0 \oplus \cdots \oplus M_n \oplus \mathfrak{a}M_n \oplus \cdots \oplus \mathfrak{a}^m M_n \oplus \cdots$$

which are finitely generated as an A^*-module because each M_n is. We have an ascending chain so $M^* = \bigcup\{M_n^*\}_{n\geq 0}$ is finitely generated iff the chain stabilizes iff $M_{n_0}^* = M^*$ for some $n_0 \geq 0$ iff $M_{n_0+n} = \mathfrak{a}^n M_{n_0}$ for all $n \geq 0$ iff the filtration is stable. □

Proposition 3.5 (Artin-Rees lemma)**.** *Let A be a Noetherian ring, M an A-module and $(M_n)_{n\geq 0}$ a stable $\mathfrak{a}$-filtration of M. Then, for any A-submodule N of M, $(N \cap M_n)_{n\geq 0}$ is a stable $\mathfrak{a}$-filtration of N. In particular, the $\mathfrak{a}$-topology of N is the restriction of the $\mathfrak{a}$-topology of M.*

Proof. It is clear that $(N \cap M_n)_{n\geq 0}$ is an $\mathfrak{a}$-filtration. So we can consider the A^*-submodule N^* of M^* which is finitely generated as M^* is. Therefore, by lemma 3.4, $(N \cap M_n)_{n\geq 0}$ is stable. □

The Artin-Rees lemma is most commonly stated as the following

Corollary 3.6. *There exists an integer $k \geq 0$ such that for all $n \geq 0$*

$$(\mathfrak{a}^n M) \cap N = \mathfrak{a}^{n-k}((\mathfrak{a}^k M) \cap N).$$

Proof. Take $M_n = \mathfrak{a}^n M$. □

Now we can apply corollary 1.14 to an exact sequence of finitely generated A-modules where A is a Noetherian ring;

Proposition 3.7. *Let $0 \to N \to M \to P \to 0$ be an exact sequence of finitely generated A-modules and A be a Noetherian ring. For any ideal $\mathfrak{a}$ of A, the sequence of $\mathfrak{a}$-adic completions*

$$0 \longrightarrow \hat{N} \longrightarrow \hat{M} \longrightarrow \hat{P} \longrightarrow 0$$

is exact.

3.2 Krull's theorem

As we already stated, we would like to know when $M \hookrightarrow \hat{M}$. In order to answer this question we will try to characterize the kernel of $M \to \hat{M}$ which we already know it's the intersection of $(\mathfrak{a}^n M)_{n\geq 0}$. To do so we will use the tensor product.

Since we have a homomorphism $\phi_A \colon A \to \hat{A}$ we can turn $\hat{A}$ into an A-algebra and by extension of scalars the tensor product $\hat{A} \otimes_A M$ can be regarded as an $\hat{A}$-module. We get the following sequence of homomorphisms

$$\hat{A} \otimes_A M \xrightarrow{\mathrm{id}_{\hat{A}} \otimes_A \phi_M} \hat{A} \otimes_A \hat{M} \longrightarrow \hat{A} \otimes_{\hat{A}} \hat{M} \cong \hat{M}.$$

Proposition 3.8. *Let A be a ring and M a finitely generated A-module then $\hat{A} \otimes_A M \to \hat{M}$ is onto. If also A is a Noetherian ring, $\hat{A} \otimes_A M \to \hat{M}$ is an isomorphism.*

Proof. By (3) and distributive we have an isomorphism

$$\hat{A} \otimes_A A^n \cong \widehat{A^n}.$$

Since M is finitely generated we have an exact sequence

$$0 \longrightarrow N \longrightarrow A^n \longrightarrow M \longrightarrow 0$$

which gives rise to the commutative diagram

prove comm. of diag.

$$\begin{tikzcd}
 & \hat{A} \otimes_A N \arrow[r, "\iota"] \arrow[d, "\alpha"] & \hat{A} \otimes_A A^n \arrow[r, two heads, "\pi"] \arrow[d, two heads, "\beta"] & \hat{A} \otimes_A M \arrow[r] \arrow[d, "\gamma"] & 0 \\
0 \arrow[r] & \hat{N} \arrow[r, "\iota'"] & A^n \arrow[r, two heads, "\pi'"] & \hat{M} \arrow[r] & 0.
\end{tikzcd}$$

The first row is exact since the functor $\hat{A} \otimes_A -$ is right exact and π' is onto by 1.14. Hence γ is onto. If A is Noetherian we now that the second bottom row is exact by proposition 3.7 and, since N is finitely generated, α is onto. This implies that $\ker(\pi') = \beta[\mathrm{im}(\iota)]$. So

$$\ker(\gamma) = \pi[\ker(\gamma \circ \pi)] = \pi[\ker(\pi' \circ \beta)] = \pi[\beta^{-1}[\ker(\pi')]] = \pi[\mathrm{im}(\iota)] = 0.$$

Hence γ is one-to-one. □

Note that when A is Noetherian, the functor $\hat{A} \otimes_A - \cong \hat{}$ is exact on the category of finitely generated A-modules.

Proposition 3.9. *Let A be a Noetherian ring endowed with the $\mathfrak{a}$-adic topology, then the following hold:*

(i) $\hat{\mathfrak{a}} = \hat{A}\mathfrak{a} \cong \hat{A} \otimes_A \mathfrak{a}$,

(ii) $\widehat{\mathfrak{a}^n} = \hat{\mathfrak{a}}^n$,

(iii) $\mathfrak{a}^n/\mathfrak{a}^{n+1} \cong \hat{\mathfrak{a}}^n/\hat{\mathfrak{a}}^{n+1}$,

(iv) $\mathfrak{a} \subseteq \mathfrak{R}_{\hat{A}}$ *where* $\mathfrak{R}_{\hat{A}}$ *is the* Jacobson radical *of* $\hat{A}$[3].

Proof. $\mathfrak{a}$ is finitely generated as an A-module so $\hat{\mathfrak{a}} \cong \hat{A} \otimes_A \mathfrak{a}$ by proposition 3.8. Moreover, the image of

$$\hat{A} \otimes_A \mathfrak{a} \to \hat{\mathfrak{a}}$$

is $\hat{A}\mathfrak{a}$. Hence (i) is proved. Similarly we get

$$\widehat{\mathfrak{a}^n} = \hat{A}\mathfrak{a}^n = (\hat{A}\mathfrak{a})^n = \hat{\mathfrak{a}}^n.$$

By corollary 1.14 and (ii) we have $A/\mathfrak{a}^n \cong \hat{A}/\hat{\mathfrak{a}}^n$. Hence (iii) is proved by taking $A = \mathfrak{a}^{n-1}$. Finally, recall that $a \in \mathfrak{R}_{\hat{A}}$ iff $1 - ab$ is a unit in $\hat{A}$ for all $b \in \hat{A}$. Given $a \in \hat{\mathfrak{a}}$, Prove it?

$$(1-a)^{-1} = 1 + a + a^2 + \cdots$$

converges in $\hat{A}$ since its complete with respect to its $\hat{\mathfrak{a}}$-topology. So $1 - a$ is a unit in $\hat{A}$. Thus $\hat{\mathfrak{a}} \subseteq \mathfrak{R}_{\hat{A}}$. □

In fact, it suffices that $\mathfrak{a}$ is finitely generated. Proposition 3.9 is stating that, if $\mathfrak{a}$ is finitely generated, $\hat{A}$ is complete with respect to its $\mathfrak{a}$-topology; equivalently, the initial topology on $\hat{A}$ coincides with the $\mathfrak{a}$-topology.

Example 3.10. *One can have a local ring* $(A, \mathfrak{m})$ *such that* $\hat{A}$ *is not complete with respect to either its* $\mathfrak{m}$*-topology nor its* $\hat{\mathfrak{m}}$*-topology. See example here. In fact, one has that, given an* A*-module* M*,* $\hat{M}$ *is complete with respect to its* $\mathfrak{a}$*-topology if and only if*

$$\widehat{\mathfrak{a}^n M} := \ker(\hat{M} \to M/\mathfrak{a}^n M) = \hat{\mathfrak{a}}^n M$$

for all $n \geq 0$ *—Lemma 10.96.5.—.*

Moreover,

Corollary 3.11. *The* $\mathfrak{m}$*-adic completion of a local ring* $(A, \mathfrak{m})$ *is local with maximal ideal* $\hat{\mathfrak{m}}$*.*

Proof. We know that $\hat{A}/\hat{\mathfrak{m}} \cong A/\mathfrak{m}$ is a field so $\hat{\mathfrak{m}}$ is a maximal ideal contained in $\mathfrak{R}_{\hat{A}}$. Thus $\hat{A}$ is a local ring. □

We are ready to answer when $M \hookrightarrow \hat{M}$ i.e., when is the $\mathfrak{a}$-topology Hausdorff:

Theorem 3.12 (Krull's theorem)**.** *Let* A *be a Noetherian ring,* $\mathfrak{a}$ *an ideal and* M *a finitely generated* A*-module endowed with the* $\mathfrak{a}$*-topology. Then the kernel of the map* $M \to \hat{M}$ *can be characterized as follows*

$$\ker(M \to \hat{M}) = \{m \in M : (1+\alpha)m = 0 \text{ for some } \alpha \in \mathfrak{a}\}.$$

Proof. We already know that $\ker(M \to \hat{M}) = \bigcap_{n \geq 0} \mathfrak{a}^n M =: N$. Since the topology induced by M and the $\mathfrak{a}$-topology are equivalent on N by proposition 3.4 and since N is the intersection of all neighborhoods of 0 we have that the neighborhood of zero $\mathfrak{a}N$ can only be N, so $\mathfrak{a}N = N$. Thus there exists an $a \in \mathfrak{a}$ such that $(1-a)N = 0$. Conversely, if $(1-a)m = 0$ then $m = a^n m \in \mathfrak{a}^n M$ for all $n \geq 0$ so $m \in N$. □ prove coro 2.5

[3] The intersection of all the maximal ideals of $\hat{A}$.

Corollary 3.13. *Let A be a Noetherian ring, M a finitely generated A-module and $\mathfrak{a} \subseteq \mathfrak{R}_A$ an ideal of A. Then the $\mathfrak{a}$-topology on M is Hausdorff. In particular, if $(A, \mathfrak{m})$ is a local ring, then both the $\mathfrak{m}$-topology on M and A are Hausdorff.*

Proof. Since $\mathfrak{a} \subseteq \mathfrak{R}_A$, $1 + \mathfrak{a}$ contains no zero divisors so $\ker(M \to \hat{M}) = 0$ and apply theorem 1.6. □

We end this subsection with the following observation which is a direct consequence of Krull's theorem.

Remark. Recall that given a ring A and a subset $S \subseteq A$ multiplicatively closed, we define an equivalence relation on $A \times S$ by

$$(a, s) \sim (b, t) \iff (at - bs)s' = 0 \text{ for some } s' \in S.$$

The quotient set is denoted by $S^{-1}A$ and becomes a ring with the obvious operations called *ring of fractions*. We will denote the equivalence class $[(a, s)]$ by a/s. Categorically speaking this ring is characterized by the following universal property:

Proposition 3.14 (Universal property of the ring of fractions)**.** *For any ring homomorphism $f\colon A \to B$ such that $f[S] \subseteq B^\times$. There exists a unique $\tilde{f}\colon S^{-1}A \to B$ such that the following diagram commutes*

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, "\iota"'] & B \\ S^{-1}A \arrow[ur, "\tilde{f}"'] & \end{tikzcd} \tag{4}$$

where $\iota\colon A \to S^{-1}A$ is the map $\iota(a) = a/1$.

If we consider the subset $S := 1 + \mathfrak{a}$ then by the universal property of the ring of fractions, since

$$(1 - a)^{-1} = 1 + a + a^2 + \cdots$$

converges in $\hat{A}$ —i.e., is a unit in $\hat{A}$— there's a ring homomorphism $S^{-1}A \hookrightarrow \hat{A}$. It's injectiveness comes from the commutativity of (4) and Krull's theorem which states that $\iota\colon A \to S^{-1}A$ and $\phi\colon A \to \hat{A}$ have the same kernel. Thus we can identify $S^{-1}A$ as a subring of $\hat{A}$.

$\widehat{S^{-1}A} = \hat{A}$ and $\widehat{A_{\mathfrak{m}}} = \hat{A}_{\mathfrak{m}}$

3.3 The associated graded ring

By now we know that if A is a Noetherian local ring then $\hat{A}$ is complete, Hausdorff and local. We wish to know wether is Noetherian or not. The following definition will come in handy.

Definition 3.15. *Let A be a ring and $\mathfrak{a}$ an ideal of A. We define the following graded ring*

$$G(A) := G_{\mathfrak{a}}(A) := \bigoplus_{n \geq 0} \mathfrak{a}^n / \mathfrak{a}^{n+1}.$$

It is clear that $G(A)$ has a natural $A/\mathfrak{a}$-module structure. However with the following product $G(A)$ becomes a graded ring:

Definition 3.16. *Given $([a_n])_{n \geq 0}$ and $([b_n])_{n \geq 0}$ elements of $G(A)$, their* product, *$([c_n])_{n \geq 0}$, is defined as*

$$c_n := \sum_{k=0}^{n} a_k b_{n-k}$$

which does not depend on the choice of representatives.

Veure la relació amb l'anell de polinomis

Similarly, if M is an A-module and $(M_n)_{n\geq 0}$ an $\mathfrak{a}$-filtration. We define a graded $G(A)$-module as follows

$$G(M) := \bigoplus_{n\geq 0} M_n/M_{n+1}.$$

We shall use the notation $G_n(M) := M_n/M_{n+1}$.

Proposition 3.17. *Let A be a Noetherian ring and $\mathfrak{a}$ an ideal. Then*

(i) $G(A)$ is Noetherian

(ii) $G_{\mathfrak{a}}(A) \cong G_{\hat{\mathfrak{a}}}(\hat{A})$ as rings.

(iii) If M is a finitely generated A-module and $(M_n)_{n\geq 0}$ a stable $\mathfrak{a}$-filtration of M, then $G(M)$ is a finitely generated graded $G(A)$-module.

Proof. Let $x_1, \ldots, x_r \in A$ such that $\mathfrak{a} = (x_1, \ldots, x_r)$. Then

$$G(A) \cong (A/\mathfrak{a})[[x_1], \ldots, [x_r]]$$

where the equivalence classes are in $\mathfrak{a}/\mathfrak{a}^2$. Thus by the Hilbert basis theorem $G(A)$ is Noetherian. This proves (i). The statement (ii) is clear from proposition 3.17. To prove (iii) let n_0 be such that $M_{n+n_0} = \mathfrak{a}^n M_{n_0}$ for all $n \geq 0$. Hence $G(M)$ is generated by $\bigoplus_{n\leq n_0} G_n(M)$ over $G(A)$. Each M_n is finitely generated as an A-module, hence each $G_n(M)$ is finitely generated as an $A/\mathfrak{a}$-module. Thus $G(M)$ is finitely generated as a $G(A)$-module. □

Let H and K be *filtred groups*; i.e., topological abelian groups with fundamental systems of 0, $(H_n)_{n\geq 0}$ and $(K_n)_{n\geq 0}$ respectively. We say that $\varphi\colon H \to K$ is a *homomorphism of filtered groups* if it's a group homomorphism and $\varphi[H_n] \subseteq K_n$ for all $n \geq 0$. Consider the groups

$$G(H) := \bigoplus_{n\geq 0} H_n/H_{n+1} \quad \text{and} \quad G(K) := \bigoplus_{n\geq 0} K_n/K_{n+1}.$$

It is clear that φ induces the following group homomorphisms:

$$\hat{\varphi}\colon \hat{H} \to \hat{K} \quad \text{and} \quad G(\varphi)\colon G(H) \to G(K).$$

Proposition 3.18. *If $G(\varphi)$ is one-to-one (resp. onto), then $\hat{\varphi}$ is one-to-one (resp. onto).*

Proof. Consider the commutative diagram of exact sequences for all $n \geq 0$

$$\begin{tikzcd}
0 \arrow[r] & H_n/H_{n+1} \arrow[r] \arrow[d, "G_n(\varphi)"] & H/H_{n+1} \arrow[r] \arrow[d, "\varphi_{n+1}"] & H/H_n \arrow[r] \arrow[d, "\varphi_n"] & 0 \\
0 \arrow[r] & K_n/K_{n+1} \arrow[r] & K/K_{n+1} \arrow[r] & K/K_n \arrow[r] & 0.
\end{tikzcd}$$

The snake lemma gives the exact sequence

$$\begin{tikzcd}
0 \arrow[r] & \ker G_n(\varphi) \arrow[r] & \ker \varphi_{n+1} \arrow[r] & \ker \varphi_n \arrow[dll, rounded corners] & \\
& \operatorname{coker} G_n(\varphi) \arrow[r] & \operatorname{coker} \varphi_{n+1} \arrow[r] & \operatorname{coker} \varphi_n \arrow[r] & 0.
\end{tikzcd}$$

If $G(\varphi)$ is one-to-one (resp. onto), $G_n(\varphi)$ is also one-to-one (resp. onto). Moreover, $\ker(\varphi_0) = 0$ (resp. $\operatorname{coker}(\varphi_0) = 0$) since $H_0 = H$ and $K_0 = K$. By induction on n we prove that $\ker(\varphi_n) = 0$ (resp. $\operatorname{coker}(\varphi_n) = 0$)[4]. □

[4] We can avoid the snake lemma if we use the —Short— Five lemma.

Proposition 3.19. *Let A be a ring, $\mathfrak{a}$ an ideal of A, M an A-module with an $\mathfrak{a}$-filtration $(M_n)_{n\geq 0}$ such that $G(M)$ is a finitely generated $G(A)$-module. If A is complete and Hausdorff with respect to its $\mathfrak{a}$-topology and $\bigcap\{M_n\}_{n\geq 0} = 0$, then M is a finitely generated A-module.*

Proof. Let $\nu_1, \ldots, \nu_r$ be a set of homogeneous generators of $G(M)$ and let $k_1, \ldots, k_r$ denote their respective degrees. For all $i \in [\![1, r]\!]$ write $\nu_i = x_i + M_{k_i+1}$ where $x_i \in M_{k_i}$. Let's denote by L^i the A-module A endowed with the topology given by the $\mathfrak{a}$-filtration $(L^i_n)_{n\geq 0} := (\mathfrak{a}^{n-k_i})_{n\geq 0}$ —where $\mathfrak{a}^n = A$ if $n \leq 0$—. Consider

$$L := \bigoplus_{i\in[\![1,r]\!]} L^i \cong A^r$$

with the direct sum topology. Let $\varphi\colon L \to M$ be the A-module homomorphism defined by mapping each $e_i \in L$ to $x_i \in M$. This ensures that elements of $L_n := \bigoplus_{i\in[\![1,r]\!]} L^i_n$ get mapped to elements of M_n, hence φ is a homomorphism of filtered groups. Thus it induces a homomorphism of $G(A)$-modules $G(\varphi)\colon G(L) \to G(M)$. By construction it is onto, therefore $\hat{\varphi}$ is onto by proposition 3.18. Consider now the diagram

$$\begin{tikzcd} L \arrow[r, "\varphi"] \arrow[d, "\phi_L"'] & M \arrow[d, "\phi_M"] \\ \hat{L} \arrow[r, "\hat{\varphi}"] & \hat{M}. \end{tikzcd}$$

Since each $L^i \cong A$ has the $\mathfrak{a}$-topology and, which is complete and Hausdorff, L is complete and Hausdorff so ϕ_L is an isomorphism. On the other hand, since M is Hausdorff, ϕ_M is one-to-one. Hence

$$\operatorname{im}(\varphi) = \phi_M^{-1}[\operatorname{im}(\hat{\varphi} \circ \phi_L)] = \phi_M^{-1}[\hat{M}] = M.$$

Thus, φ is onto and $x_1, \ldots, x_r$ generate M. □

This gives a nice corollary which proves our initial question.

Corollary 3.20. *Under the same conditions as in proposition 3.19, if $G(M)$ is a Noetherian $G(A)$-module, then M is a Noetherian A-module.*

Proof. It sufficies to prove that every submodule is finitely generated. Given a submodule $N \subseteq M$ we can filter it with $(N_n)_{n\geq 0} := (N \cap M_n)_{n\geq 0}$. This give rise to monomorphisms $N_n/N_{n+1} \hookrightarrow M_n/M_{n+1}$ hence to a monomorphism $G(N) \hookrightarrow G(M)$. Since $G(M)$ is Noetherian, $G(N)$ is finitely generated. Since

$$\bigcap\{N_n\}_{n\geq 0} \subseteq \bigcap\{M_n\}_{n\geq 0} = 0$$

N is Hausdorff. Thus, by proposition 3.19, N is finitely generated. □

If we apply this corollary with the complete ring $\hat{A}$, the Hausdorff $\hat{A}$-module $M = \hat{A}$ with the $\hat{\mathfrak{a}}$-topology knowing that $G_{\mathfrak{a}}(A) \cong G_{\hat{\mathfrak{a}}}(\hat{A})$ is Noetherian —supposing A Noetherian— by proposition 3.17 we get

Theorem 3.21. *Let A be a Noetherian ring, $\mathfrak{a}$ an ideal of A, then the $\mathfrak{a}$-completion $\hat{A}$ of A is Noetherian.*

4 Regular Rings

4.1 Krull dimension

5 Witt Vectors

A Rings and Modules

A.1 Important properties

A.2 Tensor product, exact sequences and flat modules

A.3 Local rings

References

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